

LoRA and Adapters

DATA 443/643 · Week 8, note 8.3

Week 8

American University

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Ideas from Stanford CS224N 2025 L11 (LoRA: BA, merge, swap, no extra decode latency). Original slides; that course is not copied.

Lecture plan

- 1 $W + (\alpha/r) BA$
- 2 $r(d+k)$ versus dk
- 3 Init $B = 0$, then merge or swap
- 4 Ship adapters, not documents

Key goal: one adapter per skill so XOR never has to move.

First time: what / why / formula

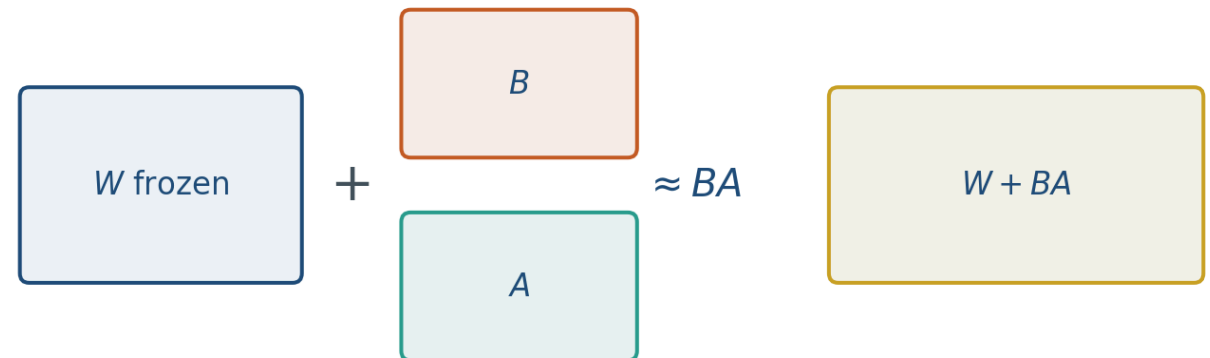
- What / why. $\Delta W \approx BA$ with rank $r \ll \min(d, k)$. Store megabytes per task, not a second 7B. Full fine-tune copies every weight (GPT-3: 175B extra per task) and overwrites (note 8.2).
- Architecture. Insert A, B on attention and/or MLP projections. Init $B = 0$ so you start as the base. QLoRA: 4-bit W , float adapters.
- Formula. $h = Wx + (\alpha / r)BAx$. Trainable count per matrix $r(d + k)$, not dk .
- Tradeoffs. + Cheap, mergeable, one adapter per skill. – Small r may fail hard domain shifts; adapters can leak in federated settings; not a new architecture.

How. Train only A, B . Merge $W \leftarrow W + (\alpha / r)BA$ at deploy (no extra latency) or swap adapters.

Freeze W , train a thin pair

- $A \in \mathbb{R}^{r \times k}$, $B \in \mathbb{R}^{d \times r}$, $r \ll \min(d, k)$.
- Usual inserts: attention projections and MLP maps.
- QLoRA quantizes W to 4-bit and still trains float adapters.

LoRA: train a thin pair BA , leave W alone



The forward map

$$h = Wx + \frac{\alpha}{r}BAx$$

Init $B = 0$ so step 0 is the base model.

Gaussian A, zero B: the first step can still move.

Trainable count is $r(d + k)$, not r^2 .

A 4096 map, rank 8

- Full W: $4096^2 \approx 16.8$ million.
- LoRA: $8 \times 8192 = 65,536$ (about 0.39%).
- A full GPT-3 copy is 175B extra per task.

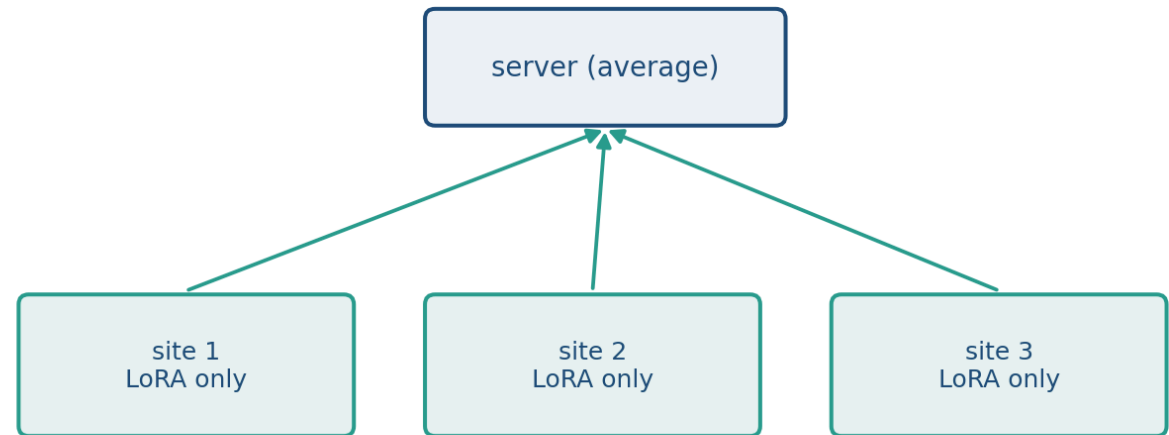
CS224N L11: a full copy of W is 175B extra for GPT-3. LoRA is this table.

$d = k = 4096, r = 8$	count
dk	$4096^2 = 16.8\text{M}$
$r(d + k)$	$8 \cdot 8192 = 65,536$
ratio	0.39%

Merge or keep a file

- Merge $W \leftarrow W + (\alpha / r)BA$: no extra decode latency.
- Swap: subtract one BA, add another.
- Federated sketch: sites send A,B. Not a privacy proof.

Federated sketch: ship adapters, not raw documents



What to take from this lecture

Count

$r(d+k)$ per adapted matrix.

Init

Zero B. Both factors zero is a deadlock.

Lab 8

4×4 printout, then XOR-then-AND without moving W.